A PROOF OF THE SUN–HOU CONJECTURE ON CIRCULAR ORDERINGS OF FINITE FIELDS WITH PRIMITIVE ADJACENT SUMS

ANONYMOUS

Abstract. Sun and Hou conjectured in 2013 that for every finite field $\mathbb{F}_q$ with $q > 7$ the elements of $\mathbb{F}_q$ can be arranged in a circle so that the sum of any two neighbours is a primitive element. Equivalently, the primitive-sum graph H_q is Hamiltonian. She recently proved this for $q > 18\,888\,871$ and for prime fields with $p > 61$. We complete the proof for all $q > 7$. A sieved Chvátal–Erdős criterion settles all but 53 of the remaining 728 fields. For these, Hamilton cycles were found by computer and checked by an independent verifier.

1. Introduction

Let H_q be the graph on $\mathbb{F}_q$ in which distinct x, y are adjacent when $x + y$ is a primitive element of $\mathbb{F}_q$. The conjecture is dated 5 September 2013; its locator in arXiv v9 is [6, Conjecture 3.5(i)].

Conjecture 1 (Sun and Hou)**.** *For every finite field $\mathbb{F}_q$ with $q > 7$ there is a numbering $a_1, \dots, a_q$ of the elements of $\mathbb{F}_q$ such that $a_1 + a_2, \dots, a_{q-1} + a_q, a_q + a_1$ are all primitive. That is, H_q is Hamiltonian.*

She [5, Theorem 1.2(1) and Remark] proved that H_q is Hamiltonian for every prime power $q > 18\,888\,871$, and for every prime $p > 61$ by a direct construction. That construction uses the theorem of Cohen, Oliveira e Silva and Trudgian on three consecutive primitive elements [3]. Luo and She [4] record the conjecture as confirmed for $q > 18\,888\,871$.

Theorem 2. *For every prime power $q > 7$ the graph H_q is Hamiltonian. Hence the Sun–Hou conjecture holds.*

2. A sieved criterion

Write $\alpha(G)$ for the independence number and $\kappa(G)$ for the vertex connectivity. All multiplicative characters, including the principal character, are extended by zero at zero. The pair-count estimate below controls both independent sets and small separated components. Applying it directly to $I \times I$, including the diagonal, improves the disjoint-partition estimate. Let

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$n = q-1$, $\theta(m) = \varphi(m)/m$, $W(m) = 2^{\omega(m)}$. For $m \mid n$, a nonzero z is *m-free* if $z = y^d$ with $d \mid m$ forces $d = 1$; primitive means n-free. For $X, Y \subseteq \mathbb{F}_q$ let $E_m(X,Y)$ count the ordered pairs $(x,y) \in X \times Y$ (with $x = y$ allowed) for which $x+y$ is m-free, and write $E = E_n$.

Lemma 3. *For a nonprincipal multiplicative character χ (with $\chi(0) = 0$) and $X, Y \subseteq \mathbb{F}_q$, $\bigl|\sum_{x\in X, y\in Y} \chi(x+y)\bigr| \le \sqrt{q|X||Y|}$.*

Proof. By Cauchy–Schwarz the square of the sum is at most

$$|X| \sum_{x\in\mathbb{F}_q} \Bigl|\sum_{y\in Y} \chi(x+y)\Bigr|^2 = |X| \sum_{y_1,y_2\in Y} \sum_{x} \chi(x+y_1)\overline{\chi(x+y_2)}.$$

The inner sum is $q-1$ if $y_1 = y_2$ and -1 otherwise, since $\sum_{t\neq 1} \chi(t) = -1$. Hence the bound is $|X|(q|Y| - |Y|^2) \le q|X||Y|$. □

Lemma 4. *Let $e \mid n$ be squarefree, let $p_1, \dots, p_s$ be the primes dividing n but not e, and suppose $\delta = 1 - \sum 1/p_i > 0$. Put $\Delta = (s-1)/\delta + 2$ if $s \ge 1$ and $\Delta = 1$ if $s = 0$. Then $E(X,Y) \ge \theta(e)\delta\bigl(|X||Y| - W(e)\Delta\sqrt{q|X||Y|}\bigr)$.*

Proof. We adapt the cancellation and prime-sieving mechanism of [3, Lemmas 4–5] to the sum condition. The character-expansion and connectivity strategy is related to [5, 4]; the shifted-product graph of [4] is different. For squarefree $m \mid n$ and $z \neq 0$,

$$\mathbf{1}_{m\text{-free}}(z) = \theta(m) \sum_{d\mid m} \frac{\mu(d)}{\varphi(d)} \sum_{\operatorname{ord}\chi = d} \chi(z).$$

Put $P = |X||Y|$ and $L = \sqrt{qP}$. The principal term is $\theta(m)(P - T_0)$, where $T_0 = \#\{x+y=0\} \le \sqrt{P}$. Lemma 3 bounds each nonprincipal character sum by L; thus $E_e \ge \theta(e)(P - W(e)L)$. If $s = 0$, then $e = \operatorname{rad}(n)$, e-free means primitive, and $\delta = \Delta = 1$, proving the assertion directly. For $s \ge 1$, the pointwise sieve inequality gives

$$E \ge \sum_i (E_{ep_i} - \theta(p_i)E_e) + \delta E_e.$$

Indeed an e-free element failing k omitted prime conditions contributes $1-k$ on the right, no more than its primitive indicator. Each bracket cancels the principal term and all orders dividing e; its absolute value is at most $\theta(e)\theta(p_i)W(e)L$. Finally $\delta + \sum_i \theta(p_i) = s - 1 + 2\delta = \delta\Delta$. □

Theorem 5. *If for some e as in Lemma 4, with $M = W(e)\Delta\sqrt{q}$ and $c = 1/(\theta(e)\delta)$, we have $\varphi(n) - M \ge M + c$, then H_q is Hamiltonian.*

Proof. As y runs over $\mathbb{F}_q$, $x+y$ runs over $\mathbb{F}_q$, so $\deg x \ge \varphi(n) - 1$; the loss is the loop $2x$. If I is a nonempty independent set, then $E(I,I) \le |I|$, so Lemma 4 gives $|I| \le M + c$. If S is a minimum vertex cut (possibly empty), A a smallest component of $H_q - S$ and B the rest, then $E(A,B) = 0$ and $|A| \le |B|$ give $|A| \le \sqrt{|A||B|} \le M$. Every vertex of A then has at least

$\varphi(n) - 1 - (|A| - 1)$ neighbours in S, so $\kappa \geq \varphi(n) - M \geq M + c \geq \alpha$. The Chvátal–Erdős theorem [1] completes the proof. □

Corollary 6 (Vertex-deletion robustness)**.** *Let $t \geq 0$ be an integer with $q - t \geq 3$. If, for an admissible e, $\varphi(q-1) \geq 2M + c + t$, then deleting any at most t vertices of H_q leaves a Hamiltonian graph.*

Proof. Delete $d \leq t$ vertices and call the remaining graph G. Its minimum degree is at least $\varphi(q-1)-1-d$, and the preceding pair-count argument still gives $\alpha(G) \leq M + c$. If G is complete the result is immediate. Otherwise apply the same argument to a minimum cut of G: its smallest component has size at most M, so $\kappa(G) \geq \varphi(q-1) - d - M \geq M + c$. Apply [1]. □

This is a direct consequence of the criterion, not a separate optimal fault-tolerance result. For example $q = 32$, $e = 1$ gives $\delta = 30/31$, $\Delta = 2$, $M = 2\sqrt{32}$ and $c = 31/30$, permitting $t = 6$.

3. Proof of Theorem 2

By [5] it remains to treat the 728 prime powers $8 \leq q \leq 18\,888\,871$ that are proper prime powers or primes $p \leq 61$. An exact computation over all choices of e (products of subsets of the primes dividing n) shows that Theorem 5 applies to all but 53 of them. The largest exception is $q = 32\,761 = 181^2$. Two independently written programs agree on this list: one uses exact rationals, the other integers only with a different enumeration. For each of the 53 fields a Hamilton cycle of H_q was found by a randomized rotation–extension search. Each cycle was checked by an independent verifier. The verifier checks $q = p^f$, certifies the field modulus by the order of X, checks the permutation property, and tests every adjacent sum for primitivity by $z^{(q-1)/r} \neq 1$, using its own polynomial arithmetic. □

For clarity about the finite-field certificate, for $f > 1$ the verifier works in $R = \mathbb{F}_p[X]/(h)$, where the recorded polynomial h is monic of degree f. This ring has exactly $p^f = q$ elements even before irreducibility is known. The tests $X^{q-1} = 1$ and $X^{(q-1)/r} \neq 1$ for each prime $r \mid q - 1$ establish that X has order $q - 1$. Its powers therefore exhaust the $q - 1$ nonzero elements of R, all of which are units. Thus R is a field and h is irreducible. Prime fields use ordinary arithmetic modulo the independently checked prime p instead.

These are not 53 previously unknown individual cases: Sun already records checks for primes below 545 and an explicit $\mathbb{F}_9$ ordering [6, Remark 3.5(a)]. The contribution is a complete, auditable closure of the finite remainder, conditional only on the cited mathematical theorems, not a claim of novelty for every certificate.

4. A classical construction in characteristic two

The primitive normal basis theorem (see [2, Theorem 1.2]) supplies a primitive element $a \in \mathbb{F}_{2^f}$ whose conjugates $a, a^2, \ldots, a^{2^{f-1}}$ form an $\mathbb{F}_2$-basis. Each conjugate is primitive, since 2 is coprime to $2^f - 1$. A cyclic binary Gray

ordering of the 2^f coefficient vectors, for $f \geq 2$, changes one coordinate at each step, including the closing step. Hence consecutive field elements sum to one of these primitive basis vectors. Such an ordering follows inductively by listing $0G$ followed by $1G$ in reverse order; the joining and closing edges change only the new coordinate. The base case is $00, 01, 11, 10$. This gives a non-search construction for every even $q > 7$. It is a classical simplification, not an additional novelty claim. The original 53 certificates are retained so that the finite closure remains independently inspectable without this alternative construction.

Appendix A. Computations

`criterion.py` and `criterion_route2.py` produce the residual list. The latter uses integer inequalities with denominators cleared; the former uses rational arithmetic. `hamsum.c` is the search and `verifysum.c` a separately implemented verifier, not an external reproduction or a formal proof assistant. The non-destructive `replay.sh` checks the frozen cycles against the explicit field moduli and hashes in `data/field-manifest.json`; optional `--regenerate` runs the search in a temporary directory and compares its output bytes. Five negative controls must be rejected. Criterion witnesses are stored separately, with exact rational slack after a nonnegative-side check before squaring.

Criterion	Covered	Residual	Largest residual
She	638	90	2 042 041
Unsieved improvement	662	66	326 041
Sieved improvement	675	53	32 761

The common scope comprises 14 primes, 592 squares and 122 higher proper powers. The sieved criterion covers respectively 0, 567 and 108 of these. As a small inspectable example, in $\mathbb{F}_{11}$ the cycle $(4, 9, 10, 8, 0, 2, 6, 7, 1, 5, 3)$ has cyclic sums $(2, 8, 7, 8, 2, 8, 2, 8, 6, 8, 7)$, all primitive modulo 11.

References

[1] V. Chvátal and P. Erdős, A note on Hamiltonian circuits, *Discrete Math.* 2 (1972), 111–113. DOI: 10.1016/0012-365X(72)90079-9.

[2] S. D. Cohen and S. Huczynska, The strong primitive normal basis theorem, *Acta Arith.* 143 (2010), 299–332. DOI: 10.4064/aa143-4-1.

[3] S. D. Cohen, T. Oliveira e Silva and T. Trudgian, On consecutive primitive elements in a finite field, *Bull. Lond. Math. Soc.* 47 (2015), 418–426. arXiv:1410.6210; DOI: 10.1112/blms/bdv018.

[4] X.-Q. Luo and Y.-F. She, A conjecture on circular permutations over finite fields, arXiv:2610.06289v1, 2026.

[5] Y.-F. She, Hamiltonicity in graphs defined by primes and primitive elements, arXiv:2609.19114v1, 2026.

[6] Z.-W. Sun, Some new problems in additive combinatorics, *Nanjing Univ. J. Math. Biquarterly* 36 (2019), 134–155; arXiv:1309.1679v9.